

LUMINOUS CAMPUS

PRODUCTION SPECIFICATION

Platform Architecture, Security System & Academic ERP Overview

Target: Next.js 16 + React 19 + Supabase + Gemini
Date: August 2026

Executive Summary: Luminous is an enterprise campus operating system uniting **high-stakes safety operations** (2-tier SOS, real-time CAD vector map, SLA timers, AI incident triage) with **complete academic ERP governance** (21 modules spanning courses, attendance, exams, hostels, placements, and wellbeing).

1 Frontend & Backend Technology Stack

 **Frontend Technology Stack**

- **Framework:** Next.js 16.3.1 (App Router, Turbopack, SSR/SSG)
- **UI Engine:** React 19.2.8 & TypeScript 5 (`strict: true`)
- **Styling:** Tailwind CSS v4.3.3 (CSS-first `@import` design tokens)
- **Component Layer:** Native React 19 primitives (zero Radix bloat)
- **Visualizations:** Recharts 3.10 + Custom SVG CAD Map
- **Iconography:** Lucide React (800+ consistent icons)
- **State:** 4 Domain Contexts (Auth, Academic, Safety, Services)

 **Backend & Data Infrastructure**

- **API Routing:** Next.js Route Handlers + Edge Proxy (`proxy.ts`)
- **Database & Auth:** Supabase (PostgreSQL with RLS & JWTs)
- **SSR Session Auth:** `@supabase/ssr` (Cookie session synchronization)
- **AI Intelligence:** Google Gemini 2.0/3.7 Flash + Heuristics Fallback
- **Validation:** Zod v4.4 schemas on all routes and AI outputs
- **Security Engine:** In-memory sliding rate limiters, CSRF, crypto IDs

2 Stack Justification: Why These vs. Alternatives

Technology	Alternatives Rejected	Core Rationale & Advantage
Next.js 16	Vite, CRA, Remix, Nuxt	Unifies SSR, Edge Proxy, Route Handlers, and React 19 in one project. CRA is dead; Vite lacks native Edge proxy middleware.
Supabase (PostgreSQL)	Firebase (Firestore), AWS Amplify	Relational schema is mandatory for ERP joins (Student $\rightarrow$ Course $\rightarrow$ Attendance $\rightarrow$ Incident). Firebase NoSQL cannot handle multi-table academic queries.
Gemini 2.0/3.7 Flash	OpenAI GPT-4o, Claude 3.5	Sub-200ms emergency triage latency at ~1/20th token cost. Paired with deterministic rule fallback for zero-downtime safety compliance.
Tailwind CSS v4	Bootstrap 5, MUI, Emotion	Zero runtime CSS overhead (~10KB purged bundle). Unlocks complete custom branding (Luminous Gold & Frosted Silver) without framework overrides.
Native Primitives	Radix UI, Ant Design, Chakra	Custom React 19 primitives save ~200KB bundle weight while maintaining full accessibility and eliminating external library vulnerabilities.

3 Security & Emergency Safety System (In-Depth)

High-Stakes Campus Safety: The safety architecture guarantees immediate emergency distress reach, automated SLA escalation, AI-assisted triage, and zero-trust data protection.

Seven-Layer Defense-in-Depth Security Pipeline

Perimeter & Network Defense

- Layer 1 (Edge Headers):** CSP, HSTS (`max-age=2yr`), X-Frame SAMEORIGIN, Permissions-Policy (GPS allowed, Mic blocked).
- Layer 2 (CSRF Guard):** Rejects cross-origin fetch (`sec-fetch-site`) & validates Origin/Referer.
- Layer 3 (Rate Limiter):** Sliding window limits: SOS (10/min), Alerts (5/min), Copilot (25/min), Auth (20/min).

Identity, Data & Payload Security

- Layer 4 (Auth Guard):** Supabase Bearer JWT & cookie session hydration.
- Layer 5 (RBAC Gateway):** 9 roles; fail-secure deny-by-default routing.
- Layer 6 (Zod Schemas):** Validates payloads and regex-sanitizes URL inputs.
- Layer 7 (Crypto IDs):** `crypto.randomUUID()` non-enumerable resource IDs.

Emergency Response & Safety Features

Safety Feature	Operational Mechanism	User Benefit & Emergency Impact
Two-Tier SOS Beacon <code>/safety/sos</code>	Level 1 (Campus): 3 taps dispatches nearest guard + starts 5-min SLA timer. Level 2 (Police): 5 taps/hold escalates to multi-squad and police.	Guaranteed instant response; automatic escalation if guards do not resolve within 5 minutes. Live GPS broadcast.
Emergency Broadcasts <code>/safety/emergency</code>	Role-gated dispatches scoped by: <i>Campus-wide, Building, Hostel Block, or Department.</i>	Eliminates mass panic by targeting only affected zones during local fires, lab leaks, or weather hazards.
AI Incident Triage <code>/api/ai/classify</code>	Gemini Flash classifies severity (LOW \$ ightarrow\$ CRITICAL), summary, and assigns response unit with deterministic fallback.	Lab smoke/hazard automatically triggers immediate electrical isolation and Level 1 evacuation directives in milliseconds.
Command Center & CAD Map <code>/campus-map</code>	Interactive SVG architectural vector blueprint with pulsing hazard pins, patrol units, and facility inspection.	Gives security officers single-pane spatial awareness of active campus hazards and nearest available guards.
FERPA-Compliant Copilot <code>/copilot</code>	Zero-trust tool authorization. Students see only own data; Parents see only linked child; Whistleblower IDs redacted.	Natural language assistance without exposing confidential student grades, records, or reporter identities.

4 ERP Platform Architecture & Institutional Benefits

Luminous is a unified **21-Module Higher Education ERP** that eliminates disconnected point solutions.

Academic Governance (7 Modules)

- **Departments:** Budget allocations, lab counts, HOD governance.
- **Courses & Syllabi:** Credits, semesters, instructor allocations.
- **Faculty Management:** Profiles, cabins, research, teaching hub.
- **Classroom Attendance:** Batch marking, defaulter alerts (<75%).
- **Examinations:** Mid-Term/Final scheduling, SGPA analytics.
- **Master Timetable:** Grid/Agenda views with room clash prevention.

Student & Campus Operations (7 Modules)

- **Student Directory & Portal:** Roster, CGPA, daily schedule.
- **Parent Cockpit:** Ward attendance, grades, curfew visibility.
- **Hostel & Curfew:** Room allocations, maintenance, night logs.
- **Placement Services:** Corporate drives, auto-CGPA eligibility.
- **AI Grievance Redressal:** 8-domain auto-ticket routing.
- **Student Wellbeing:** Confidential counseling & crisis lines.

The ERP Cross-Module Advantage (Zero Data Silos)

- **Exams $\rightarrow$ Placements:** Published CGPA and backlogs in `/exams` automatically lock/unlock eligibility for company drives in `/placement`.
- **Attendance $\rightarrow$ Parent Portal:** Faculty submissions in `/attendance` immediately reflect in `/parent` for attendance defaulter warnings.
- **Hostel $\rightarrow$ Security:** Curfew turnstile anomalies in `/hostel` immediately trigger patrol alerts in `/security`.
- **Safety $\rightarrow$ Interactive Map:** Reports filed in `/incidents` project live hazard beacons onto the vector CAD blueprint in `/campus-map`.

5 Future Expansion Roadmap: What Else Can We Do?

Near-Term (Phase 1)

- **Web Push / PWA:** Emergency push alerts even when browser is closed.
- **Dark Mode UI:** Night-shift theme for security desk operations.
- **PDF/CSV Export:** Attendance sheets, grade cards, audit logs.
- **SMS/Email Fallback:** Twilio SMS for critical campus emergencies.

Medium-Term (Phase 2)

- **Mobile Native Apps:** React Native / Flutter apps with offline SOS.
- **CCTV Camera Stream:** Real-time feeds in Command Center with AI anomaly alerts.
- **Tuition & Fee Gateway:** Payment reconciliation and scholarship tracking.
- **Fleet GPS:** Live campus shuttle tracking and QR transport passes.

Long-Term Vision (Phase 3)

- **Smart Building IoT:** Smoke/gas detectors feeding directly into Safety Context.
- **Multi-Campus Federation:** Centralized governance across university branches.
- **AI Vision Surveillance:** Automated perimeter breach & crowd analytics.
- **Voice-Activated SOS:** Hands-free voice emergency trigger ("Hey Luminous").